CareSpeak: A Conversational Training Tool for APRN Students

Challenge Overview

Nursing students need realistic opportunities to practice sensitive clinical communication. Traditional standardized patient encounters are costly and difficult to scale. There is a clear need for accessible, emotionally realistic AI-based training tools to support competency-based education in a scalable, accessible, and repeatable way.

AI-Based Solution

CareSpeak provides real-time, voice-based practice for sensitive clinical conversations using AI patients. Faculty can monitor sessions live and offer targeted feedback. Unlike general AI like ChatGPT, CareSpeak was developed with direct input from APRN clinicians to ensure scenarios address specific nursing communication challenges. This user-centered approach prioritizes emotional realism, contextual nuance, and learning outcomes closely aligned with faculty objectives. Inspired by research like GuideLLM, it uses clinician-curated personas to focus on goal navigation, context management, and empathetic engagement—enhancing clinical authenticity, relevance, and the likelihood of sustained adoption within nursing education.

Completed Work (April 2025)

- WebSocket-based architecture enabling real-time student-AI conversation with both synchronous and asynchronous monitoring from faculty for evaluation.
- Frontend interfaces for students and faculty, fully integrated AI speech services and peer-to-peer chat.
- Prompting system based on dynamically loaded patient personas and scenario goals.

Future Directions (Continued work for Master's thesis)

- Emotion-aware modeling using speech tone and prosody.
- Expanded scenario library featuring diverse cultural, clinical, and linguistic cases.
- Further collaboration with content experts for AI evaluation and usability testing.
- Integration with learning platforms like Canvas and Blackboard.
- (Long term) Video avatar integration to simulate telehealth and non-verbal cues.

Ethical and Legal Considerations

- All scenarios are based on synthetic patients; no real patient data is used.
- HIPAA is not directly applicable at the prototype stage, but safeguards would be necessary for any real clinical deployment.
- **FERPA considerations are critical**, especially regarding the storage and use of student performance data, feedback, or evaluations within the system.
- Commercial AI services (e.g., Hume AI) are used as part of a proof-of-concept only.
- Any future use in nursing education must ensure privacy, transparency, and compliance with institutional policies for student data protection.

CareSpeak: User Interface

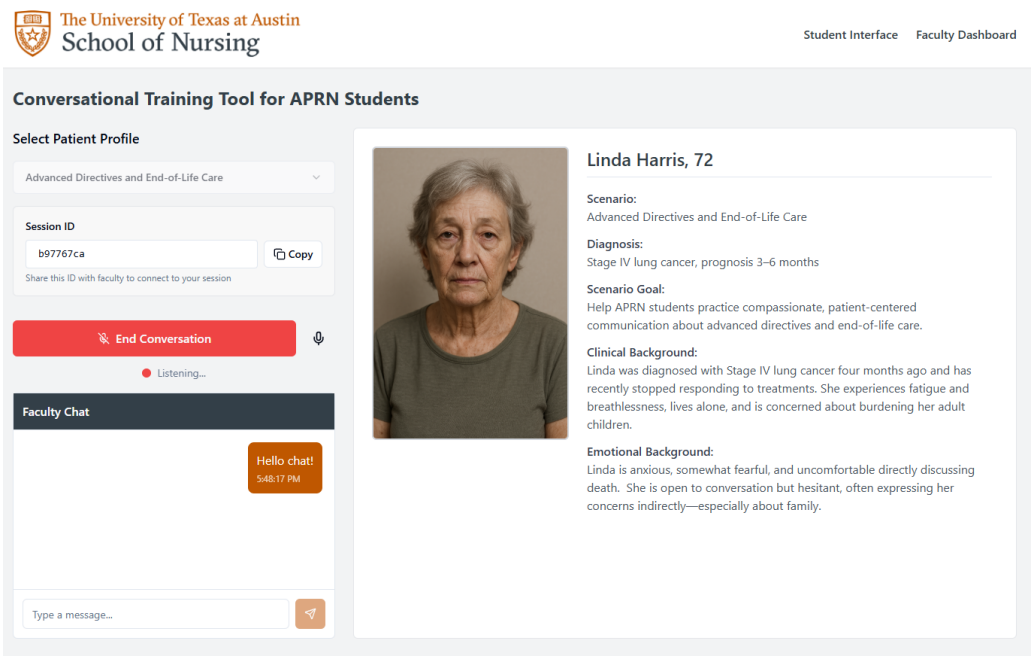


Figure 1: CareSpeak student interface, showing persona selection, session ID, and faculty chat panel.

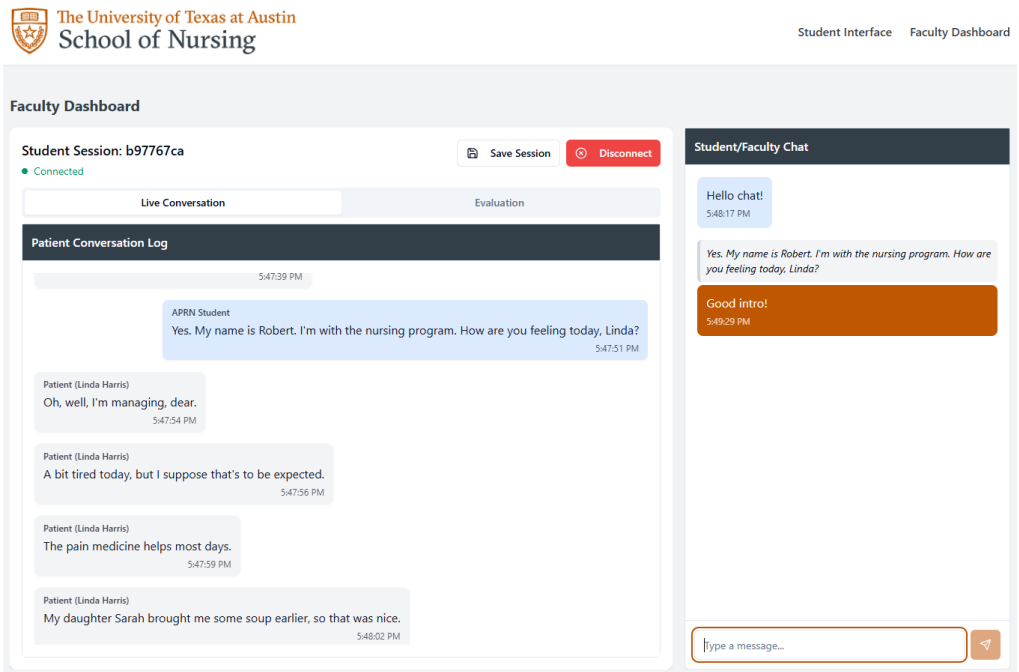


Figure 2: Faculty Dashboard view, showing the live transcript log and chat interface, demonstrating quote functionality.